A MAJOR PROJECT SYNOPSIS

ON

FabMetrics AI

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the Major Project topic "**FabMetrics AI**" submitted by **Chitransh Saxena (9923102040), Dhruv Kumar (9923102051), and Shubhangi Mathur (9923102054)** is new, appropriate and not repeated/copied from previously submitted project works.

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DECLARATION

We hereby declare that the title of the Major Project, "**FabMetrics AI**," is not repeated/copied from previously submitted project works and that we have not misrepresented, fabricated, or falsified any idea, data, fact, result, or source in our submission.

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PROBLEM STATEMENT

Semiconductor manufacturing involves testing thousands of dies on a wafer. When failures occur, they often form visible patterns such as rings, scratches, centre defects or edge-related defects. These patterns are useful because they can give engineers an indication of where a manufacturing or handling problem may be occurring. Manually studying a large number of wafer maps, however, is slow and can become inconsistent.

While exploring wafer-map classification, we noticed that every defect pattern does not look important in the same way. Patterns such as Center, Donut and Edge-Ring are easier to understand from the overall wafer shape, whereas Scratch and localized defects need more attention to smaller regions. This led to our main project idea: instead of asking one network to learn everything, we want to check whether two complementary networks can focus on different details and work better together.

To test this idea, we use a dual-branch model. ResNet-50 with CBAM looks mainly at the overall wafer structure and important regions, while EfficientNet-B0 captures finer and multi-scale details. Instead of simply joining both outputs, cross-attention is used so that the two branches can influence which information is useful before the final class is predicted.

The project uses the WM-811K wafer-map dataset as its primary data source. The dataset is highly imbalanced, so the original wafer maps are first separated into training, validation and test groups. Augmentation is then applied only to the training data to avoid data leakage. The planned training set contains 35,000 images, with 3,500 images for each of ten classes: Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, None and a controlled Multi-Defect class. Along with classification, the project provides defect-region visualization and a simple inspection dashboard so that the model can be demonstrated as a complete student-developed inspection system.

Project Idea at a Glance

OUR PROJECT IDEA	WHY WE ARE DOING IT
Main question	Can one branch learn the overall wafer pattern while another focuses on smaller details, and can combining them improve classification?
What we noticed	Some defects are mainly identified by their overall geometry, while others depend more on local or fine-grained regions.
What we decided to try	Use ResNet-50 + CBAM and EfficientNet-B0 as two complementary branches, then combine their useful features using cross-attention.
How we will check it	Compare our model with simpler baselines and remove/change one component at a time through an ablation study.

OBJECTIVES

- Prepare and organize the WM-811K wafer-map data using a source-level train/validation/test split before augmentation.
- Create a balanced training set of 35,000 images while keeping augmented copies out of validation and test data.
- Develop a dual-branch model using ResNet-50 + CBAM and EfficientNet-B0 to learn complementary wafer features.
- Combine the two feature streams using cross-attention and classify ten wafer-map categories, including a controlled Multi-Defect class.
- Compare the proposed model with simpler baseline models to check whether each added component actually improves performance.
- Evaluate the model using Macro F1-score, accuracy, precision, recall, class-wise F1-score, confusion matrix and inference latency.
- Use OpenCV contours and bounding boxes to provide defect-region visualization as an inspection aid.
- Integrate the trained model with a FastAPI-based dashboard, authentication and PDF inspection reporting.
- Perform an ablation study to understand the contribution of CBAM, the second branch, cross-attention, focal loss and SWA.
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INTRODUCTION

Wafer maps are generated during semiconductor testing to show the pass/fail status of dies across a wafer. When failures are concentrated in a particular area, their spatial arrangement can form recognizable patterns. Automatically identifying these patterns can help in faster screening and can support engineers while investigating recurring process abnormalities.

During the initial design of FabMetrics AI, our main question was simple: what if one network looks at the wafer as a whole while another pays more attention to smaller details? Based on this idea, we selected two branches. ResNet-50 with CBAM is used for the broader wafer pattern, while EfficientNet-B0 is used for finer multi-scale features. We then use cross-attention so that the branches do more than just combine outputs—they can help decide which features from the other branch are more useful.

The project considers ten output classes: Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, None and Multi-Defect. The first nine are based on wafer-map defect categories used in the project dataset. The Multi-Defect class is generated through controlled superposition for experimentation and will therefore be reported separately from genuine real-fab mixed-defect validation.

One important issue we want to avoid is data leakage. If an image is augmented first and its transformed copies are later divided between training and testing, the model may indirectly see very similar samples during training. Therefore, original source wafers are split first and augmentation is performed only on the training portion.

For us, the main project is the defect-classification model and the experiments used to check whether our design choices actually help. Focal loss is tested for difficult examples and Stochastic Weight Averaging (SWA) is evaluated as a training improvement. The OpenCV visualization, web dashboard and PDF report are supporting features that make the final prototype easier to demonstrate and understand.

Dataset Preparation Plan

DATASET PART	WHAT WE ARE USING / DOING
Primary dataset	WM-811K, a large public wafer-map dataset with 811,457 wafer maps and a highly imbalanced class distribution.
Base classes	Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full and None.
Multi-Defect experiment	We create controlled synthetic combinations for experimentation and report them separately from genuine mixed-defect wafer data.
How we split the data	Original source wafers are divided into Train / Validation / Test sets before any augmentation.
How we balance training	Augmentation is applied only to the training portion until the planned 3,500 training images per target class are reached.
How we keep evaluation fair	Validation and test sets remain source-separated and non-augmented; final frozen counts will be documented in the report.

METHODOLOGY

- **Dataset Preparation:** Load and clean WM-811K wafer maps, verify labels and record the class distribution used in the project.
- **Data Split:** Divide original source wafers into training, validation and test sets before performing augmentation.
- **Training Data Balancing:** Apply augmentation only to training samples until the planned 3,500 images per class are obtained. Generate the experimental Multi-Defect samples using a documented controlled procedure.
- **Pre-processing:** Resize wafer maps to 224 × 224 pixels and apply the same normalization pipeline to all models used for comparison.
- **Feature Extraction:** Pass each wafer map through ResNet-50 + CBAM for global/topological information and EfficientNet-B0 for fine-grained multi-scale information.
- **Feature Fusion:** Project both feature streams to a common feature space. Cross-attention uses Query (Q), Key (K) and Value (V) representations so that features from one branch can assign higher importance to useful information from the other branch: Attention(Q,K,V) = softmax(QK^T/sqrt(d))V.
- **Training and Classification:** Train the classifier using focal loss and evaluate SWA as a separate training improvement. The final layer produces one of the ten target classes with a confidence score.
- **Testing and Comparison:** Compare the proposed model with single-backbone and simpler fusion baselines using Macro F1, accuracy, precision, recall, class-wise F1, confusion matrix and latency.
- **Inspection Interface:** Display the predicted class, confidence and OpenCV-based defect-region overlays on the dashboard, with an option to generate a PDF inspection report.

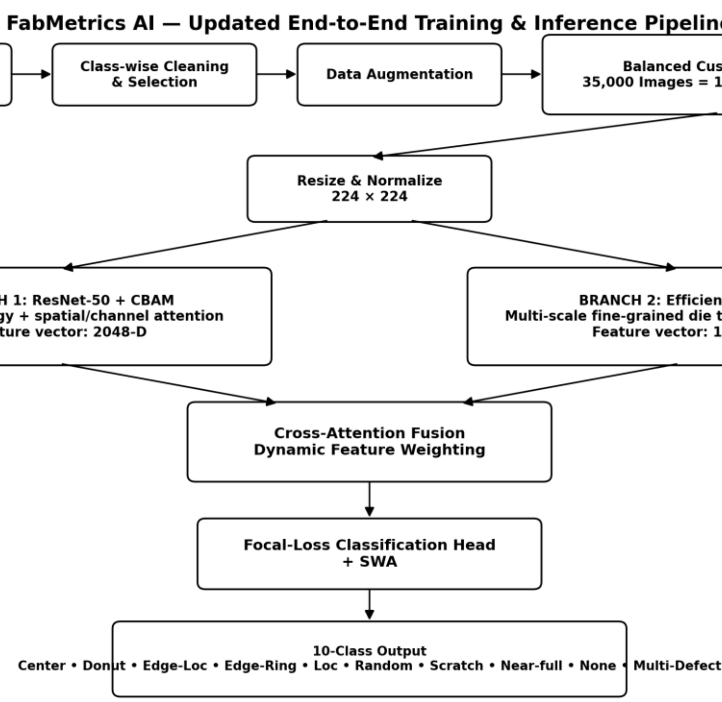

Figure 1. Proposed FabMetrics AI workflow: wafer-map preparation → dual feature extraction → cross-attention fusion → defect classification → inspection visualization.

Software/Tools: Python, PyTorch, Torchvision, OpenCV, NumPy, Pandas, Scikit-learn and FastAPI. Supporting interface/reporting tools include HTML/JavaScript/Tailwind CSS and ReportLab. Google Gemini is limited to the optional Cleanroom AI Tutor and is not part of the defect-classification model.

Evaluation Plan

WHAT WE WILL CHECK	HOW WE WILL READ THE RESULT
Main metric	Macro F1-score, because it gives equal importance to each class.
Supporting metrics	Accuracy, precision, recall and class-wise F1-score.
Where the model gets confused	Confusion matrix and observations for difficult or similar-looking classes.
Practical speed	Inference latency, reported along with the hardware configuration used for testing.

Ablation Study Plan

RUN	WHAT WE CHANGE	WHAT WE WANT TO UNDERSTAND
A1	Remove CBAM	Does channel/spatial attention help the ResNet branch?
A2	Remove EfficientNet-B0 branch	Do the additional fine-grained features actually help?
A3	Use simple concatenation instead of cross-attention	Is cross-attention better than simply joining the two feature vectors?
A4	Use cross-entropy instead of focal loss	Does focal loss help with the difficult examples?
A5	Train without SWA	Does weight averaging improve generalization?
A6	Run only the original 9 classes	Does adding the synthetic Multi-Defect class affect the original classes?

Baseline Comparison Plan

MODEL / EXPERIMENT	WHY WE ARE INCLUDING IT
ResNet-50	Gives us a straightforward single-backbone baseline for comparison.
EfficientNet-B0	Shows how the lightweight fine-feature branch performs by itself.
ResNet-50 + CBAM	Lets us check whether attention improves the ResNet baseline.
ResNet-50 + EfficientNet-B0 (concatenation)	Checks whether simply using two branches is enough without cross-attention.
Our dual-branch model + cross-attention	Checks whether learned interaction between the two branches adds value beyond simple concatenation.

IMPORTANCE OF THE PROJECT IN CONTEXT OF CURRENT SCENARIO

Semiconductor manufacturing requires fast and consistent quality inspection, and wafer maps provide a compact way of observing how failures are distributed across a wafer. As wafer volumes increase, automatically recognizing these patterns can reduce the amount of repetitive manual screening and help engineers focus on unusual cases.

FabMetrics AI is relevant because it lets us apply computer vision to a real semiconductor-inspection problem while still asking a clear engineering question. We are not adding multiple deep-learning components only to make the architecture look complex; we want to check whether global and fine-grained features actually complement each other, and whether cross-attention gives a useful improvement over simpler fusion methods.

The project also pays attention to how the dataset is prepared. Since WM-811K is highly imbalanced, balancing is performed only after the original source split. This makes the evaluation more meaningful and reduces the chance of obtaining artificially high results because of augmentation leakage. The Multi-Defect experiment further allows us to study a more difficult case while clearly stating that synthetic combinations are not a substitute for real mixed-defect wafer data.

Finally, the dashboard, region visualization and PDF report show how the trained model can be converted into a usable inspection-support prototype. These features make the project more complete for demonstration, while the main academic work remains the model design, dataset preparation, baseline comparison and ablation study.

TIME SCHEDULE OF ACTIVITIES

Work will be done by Mid Viva:

- Verify literature and finalize the research gap, hypothesis, target classes, and evaluation protocol.
- Audit WM-811K and document the exact source distribution used in the project.
- Freeze source-level train/validation/test split before augmentation.
- Implement training-only balancing and controlled Multi-Defect generation.
- Implement ResNet-50, EfficientNet-B0, and ResNet-50 + CBAM baselines.
- Run preliminary experiments and record reproducible configuration (seed, split, hardware).

Work will be done by End Viva:

- Implement and mathematically verify bidirectional cross-attention fusion.
- Complete proposed-model training and class-wise evaluation on the frozen split.
- Run all baseline and ablation experiments; compare Macro F1, class-wise F1, accuracy, and latency.
- Validate/qualify the synthetic Multi-Defect results and test real mixed-defect data if available.
- Complete OpenCV region visualization and supporting dashboard/PDF integration.
- Prepare final report with reproducible experiment settings, limitations, and evidence-backed conclusions.

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